

EVA AGUIRRE

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EDUCATION

Bachelor of Science in Biomedical Engineering

May 2026

Florida State University • GPA: 3.615

Tallahassee, FL

Master of Science in Medical Technology (MedTech)

May 2027

Lampe Joint Department of Biomedical Engineering: NC State & UNC Chapel Hill

Raleigh, NC

TECHNICAL SKILLS

Product & Design: Iterative Device Design, Design Controls, Regulatory & Quality Awareness, Manufacturing Feasibility, SOP Compliance & Documentation

Research & Analysis: Mammalian Cell Culture, Fluorescence Microscopy, Quantitative Image Analysis (FIJI, MATLAB), Statistical Analysis (Prism), Bioprinting & Rheology

Core Competencies: Experimental Workflow Optimization, Device Troubleshooting, Cross-functional Collaboration, Scientific Communication

EXPERIENCE

Biomedical Engineering Intern

Aug 2026 – present

Pulvinar Neuro

Durham, NC

- Contributed to design control matrix, ISO 14971 risk management plan, and FDA deficiency responses for a neurostimulation device.
- Performed oscilloscope testing and analyzed output waveform data to support device verification.

Senior Design Project Member – Mayo Clinic Convergence Science Internship

Aug 2025 – Apr 2026

Florida State University

Tallahassee, FL

- Led iterative device design for a therapeutic prototype targeting an unmet clinical need, integrating physician feedback from live clinical observations into successive design revisions.
- Coordinated cross-functional development spanning regulatory strategy, manufacturing feasibility, and quality assurance across a five-member team, mirroring an industry product development workflow.

Research Intern – Research Experience for Undergraduates

May 2025 – Jul 2025

Cell Manufacturing Technologies, Georgia Institute of Technology

Atlanta, GA

- Designed and validated an automated image analysis pipeline integrating FIJI and MATLAB to quantify mitochondrial internalization in high-dimensional microscopy datasets, producing a reproducible, transferable research method.
- Delivered findings in two formal presentations: an end-of-program research talk and a poster at BMES 2025.

Undergraduate Researcher

Sep 2024 – Apr 2026

National High Magnetic Field Lab & Florida State University

Tallahassee, FL

- Performed real-time troubleshooting of bioprinting and rheology instrumentation, maintaining data integrity and minimizing workflow disruption during active experiments.
- Spearheaded an independent Honors thesis project, coordinating with lab members to drive experimental design.

Volunteer Researcher

May 2023 – Aug 2023

The Miami Project to Cure Paralysis

Miami, FL

- Independently designed and executed experiments with meticulous procedural documentation, building foundational habits aligned with regulated research and quality environments.
- Executed tissue processing, staining, and microscopy to confirm Schwann cell presence in rat facial nerves.

PROJECTS (Portfolio Link)

FENNIX Medical: Physician Modified Endograft Platform

Aug 2026 – Present

- Developing a tool to cut physician-modified endograft prep time from about 6 hours to under 1 hour, in partnership with a vascular surgeon sponsor.
- Running operator reproducibility and durability bench testing, including cyclic fatigue and seal integrity, using custom ImageJ analysis and nested ANOVA.

Engineered Ileal Drug Delivery System

Aug 2025 – Present

- Designed a localized drug delivery system for ileal Crohn's disease

- Defined market segmentation and clinical entry strategy; developed full pitch materials including regulatory pathway, manufacturing feasibility, and patient access framing.

Biofabricated Muscle Tissue Model (HITM)

Aug 2025 – Apr 2026

- Developed 3D muscle tissue constructs with rheologically tuned bioinks to emulate native muscle biomechanics, independently leading scaffold design, process optimization, and experimental iteration.
- Presented research outcomes at FURC 2026 and to the thesis committee, communicating design rationale and technical results to mixed scientific audiences.

AFFILIATIONS

Honors in the Major: Undergraduate Research Program – Scholar

Phi Delta Theta Tau – Professional Fraternity Member | Community Service & Brotherhood Committees